



DV FINAL PROJECT

MG-GY 6203, DATA VISUALIZATION FOR
BUSINESS INTELLIGENCE

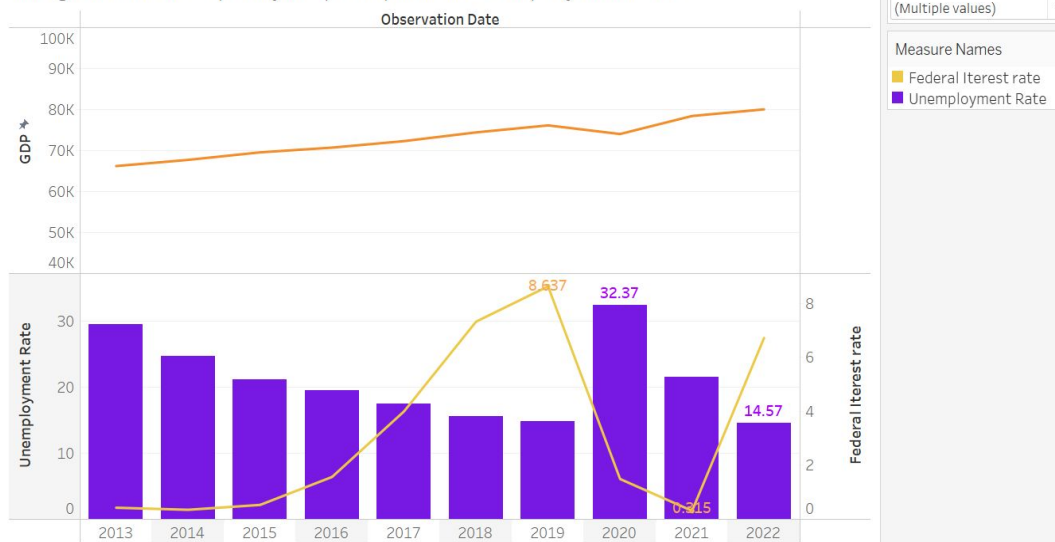
Background

The Covid-19 Pandemic caused a sharp decline in economic activity and a rise in unemployment, but government stimulus measures helped the economic recovery. The US economy has shown resilience but ongoing monitoring of key indicators is crucial for understanding the pandemic's impact and guiding policy decisions.

The turning point after pandemic

The graph shows a significant turning point after either 2019 and 2021 in this three economic statistics. However, after the turning point in 2021, the graph is showing a trend that goes against the general relationship between these interest rate, inflation and Unemployment rate.

The government's policy helps improve the employment rate



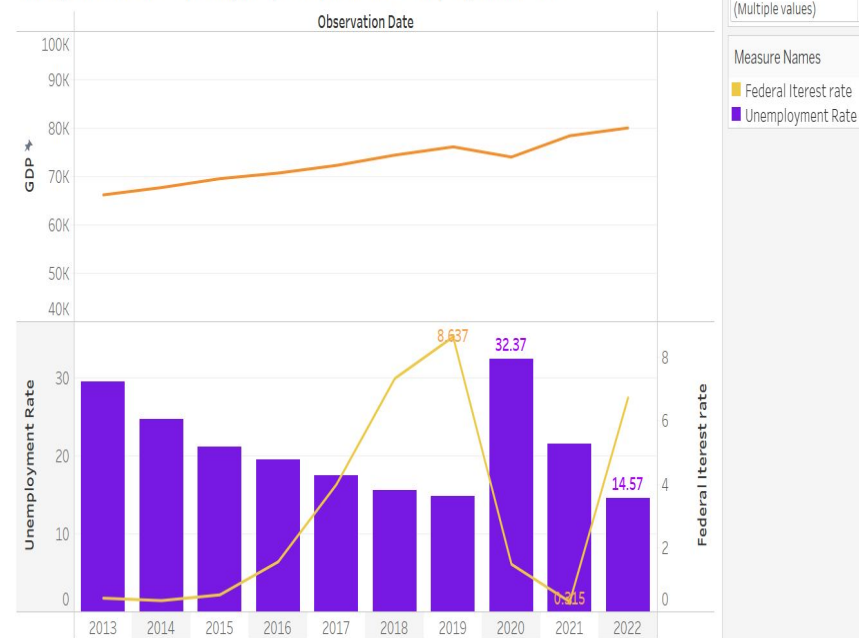
The effect of pandemic and government policy take into account

The effect of pandemic and government policy works together to form this reverse trend:

The primary reasons for the increase in interest rates in 2021 is the strong economic recovery from the COVID-19 pandemic.

In response to the pandemic, the Federal Reserve had lowered the federal funds rate to **near-zero levels** in 2020-2021 to encourage borrowing and stimulate the economy. However, as the economy began to recover and consumer demand increased, concerns about inflationary pressures began to grow.

The government's policy helps improve the employment rate



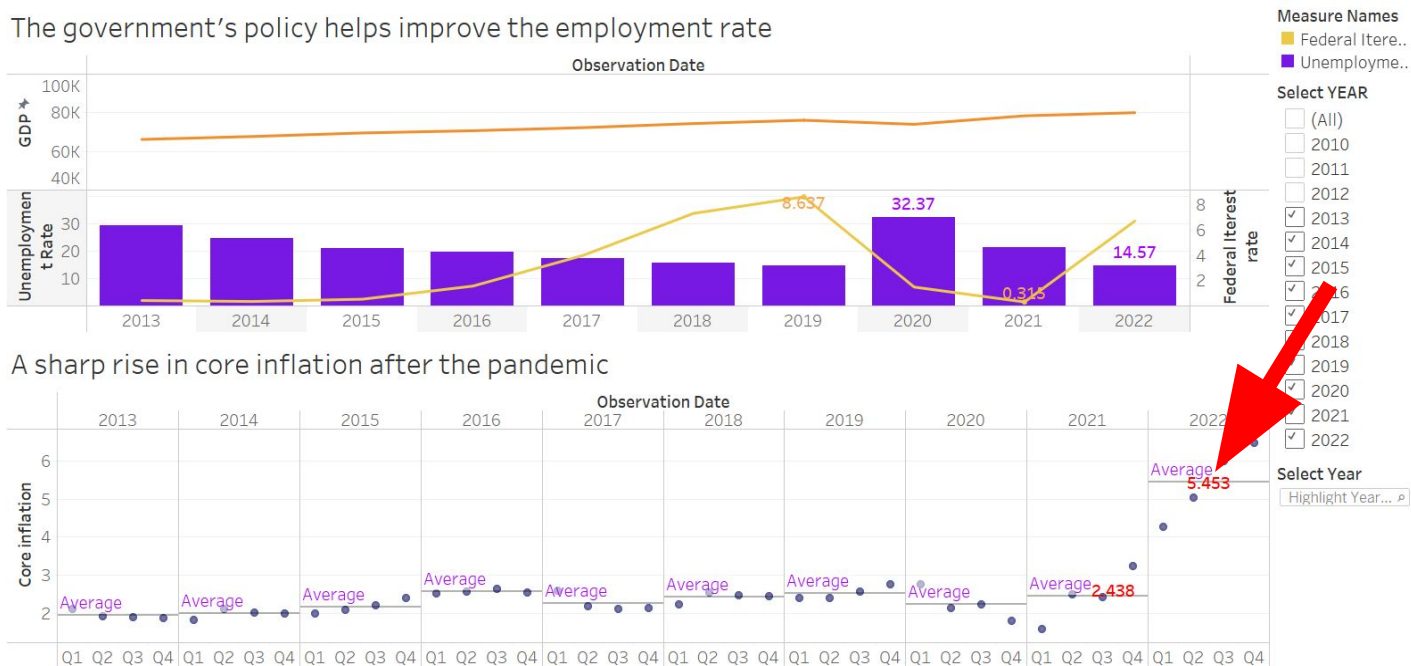
Side-effect of Government's policy

Raising interest rates to prevent inflation from getting out of control

The government's policy helps improve the employment rate

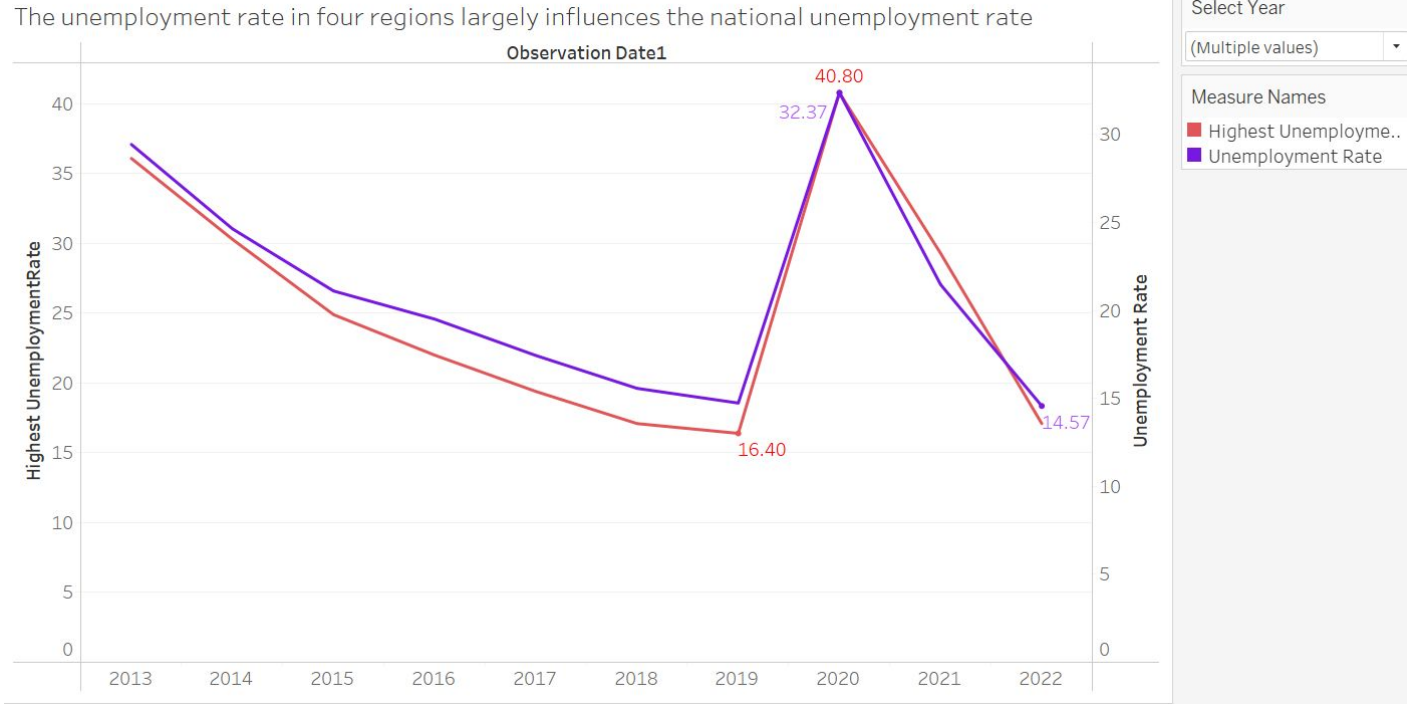
Notice:

However, the decrease of interest rate might cause the inflation go out of control.



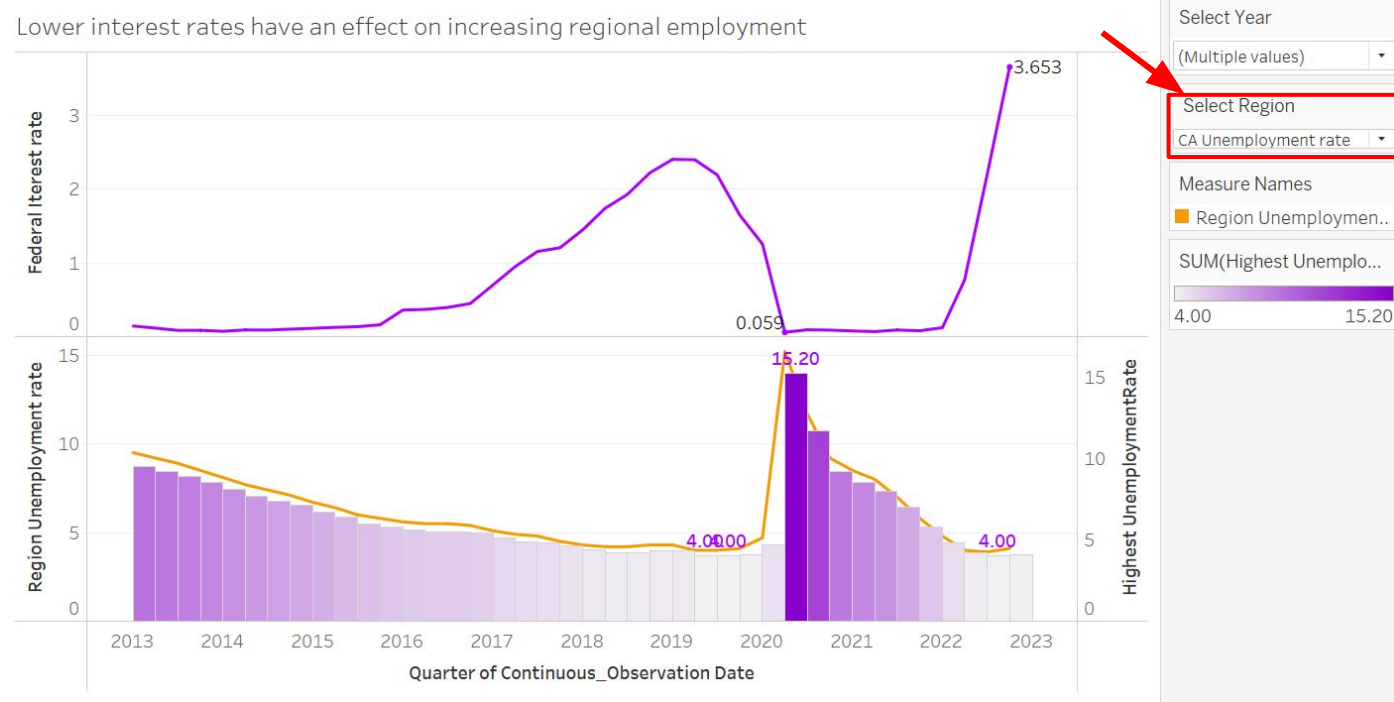
The highest unemployment rate in 4 regions

The highest unemployment rate in 4 regions shows a similar pattern to the national unemployment rate. The unemployment rate in four regions largely influences the national unemployment rate.



Deep effects on Government's policy

Lower interest rates have a positive effect on increasing regional employment. For more research in different regions, please use Tableau to select the region.

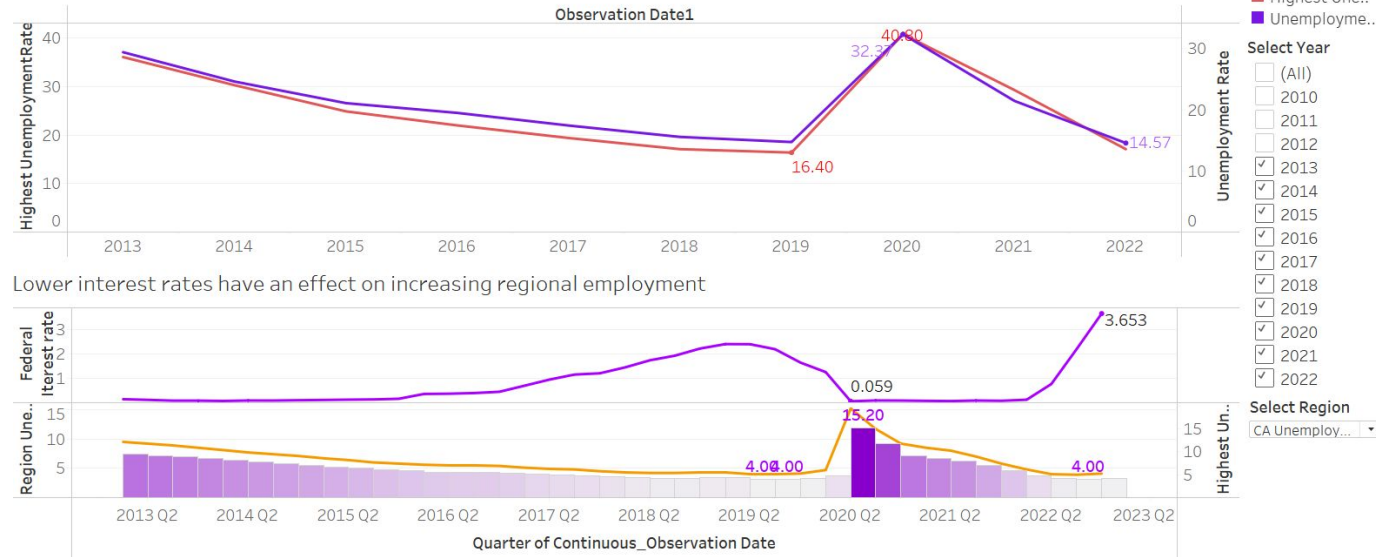


Key to focus on four regions

Strengthening economic policies in these four regions would provide a significant boost to the national economy.

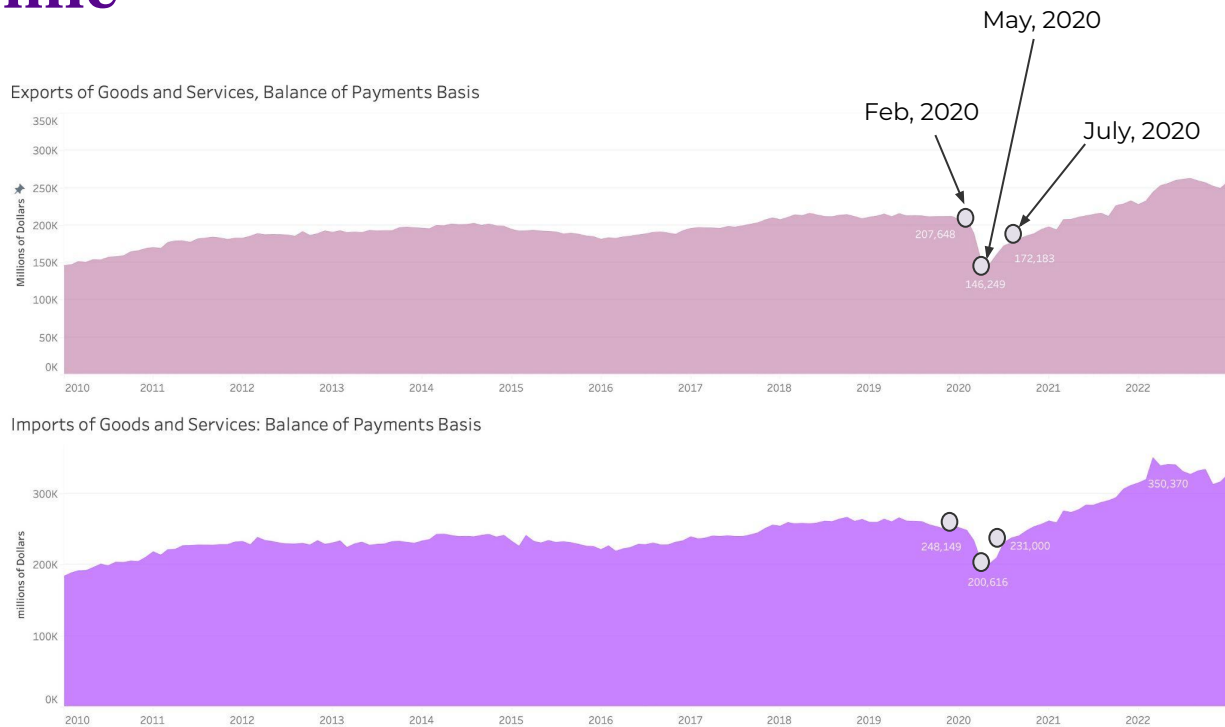
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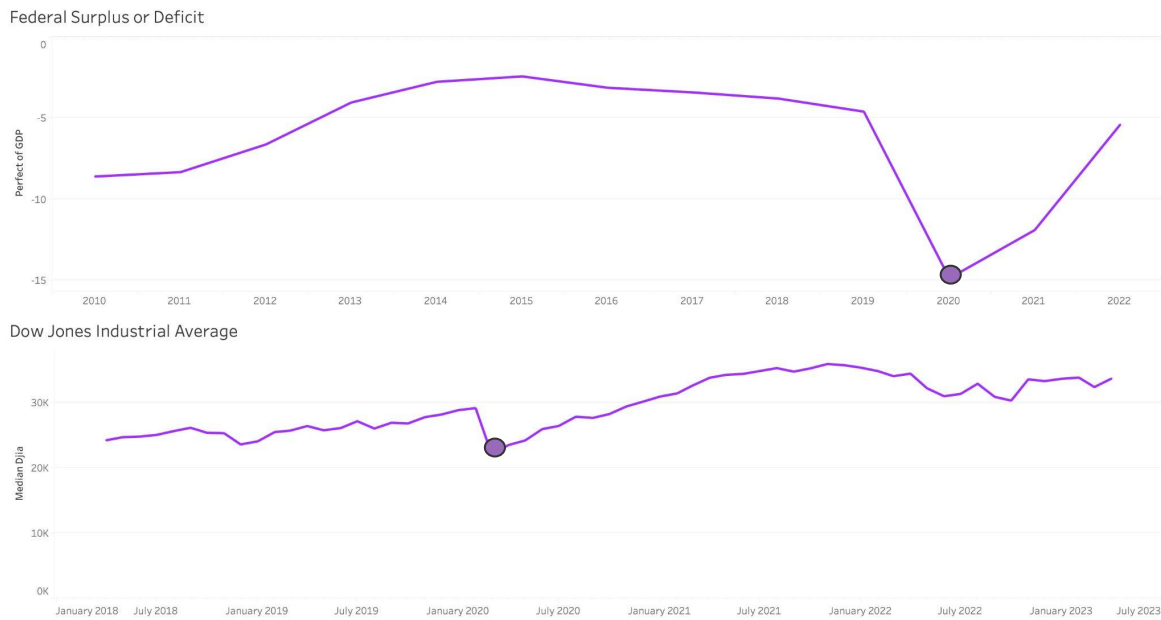
The Economy of Import and Export Due to the Impact of the Pandemic

- Economic stagnation during the pandemic
- As the pandemic stabilized, imports and exports began to grow consistently



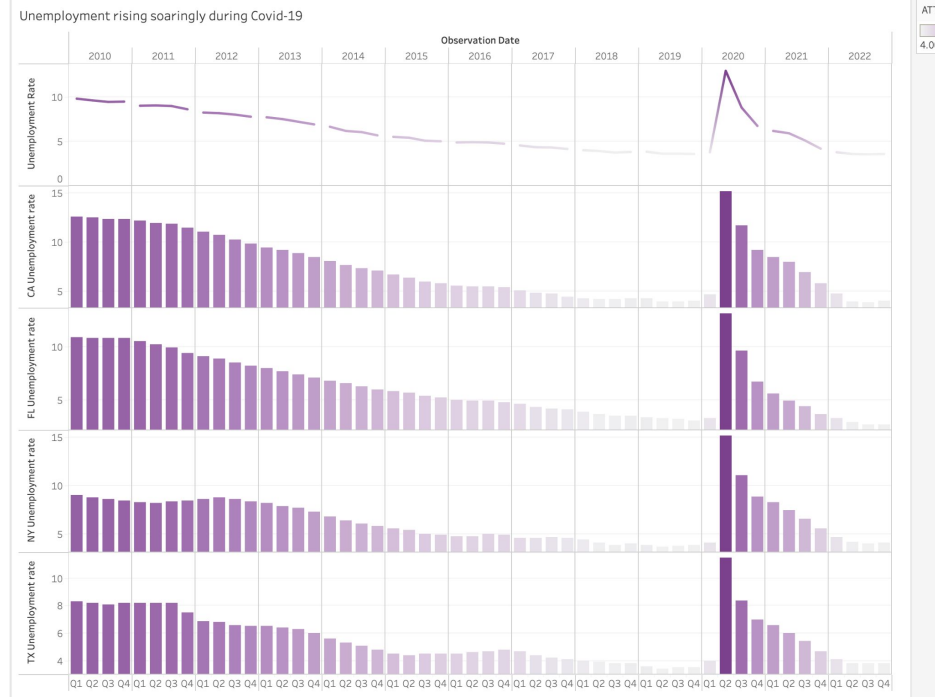
The effect of Federal Surplus or Deficit & Dow Jones Industrial Average

- Dow Jones Industrial Average shows that the stock market is bad and also affects the economy
- Federal Surplus or Deficit affects high inflation



The reason why NY, California, Texas, Florida, the four states unemployment rate fell

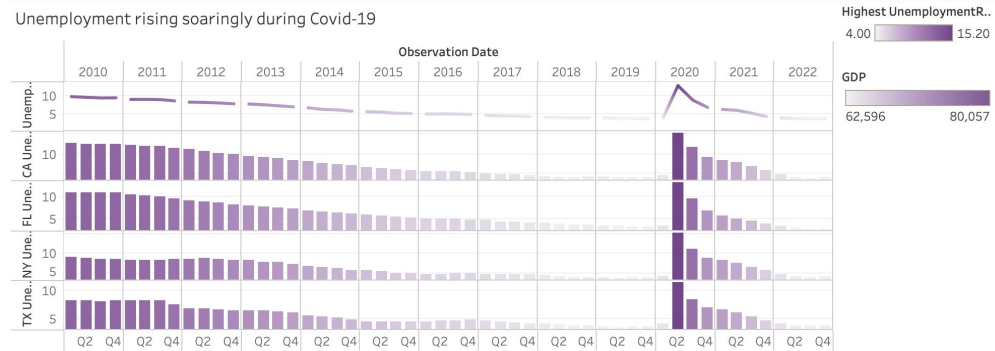
- The covid-19 ended in 2022 economy begins to recover, with unemployment falling as the businesses in states begin to reopen.



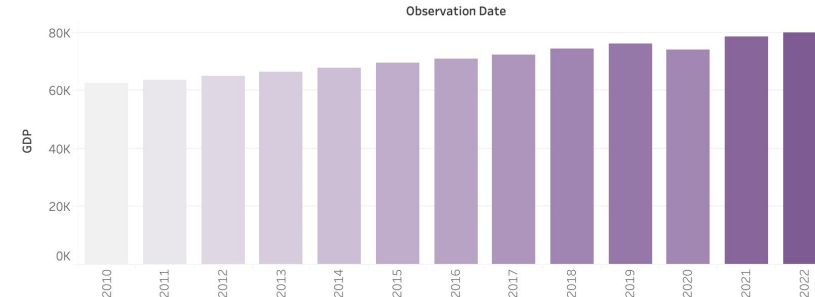
National Unemployment Rate vs State Unemployment Rate

During the covid-19 period, the U.S. gdp per capita fell and the unemployment rate reached a record high.

Unemployment rising soaringly during Covid-19



U.S. per capita gdp changes year by year



Recommendations & Takeaways

The Federal Reserve had lowered the federal funds rate to near-zero levels might cause the inflation go out of control.



Strengthening economic policies in these four regions would provide a significant boost to the national economy.



With the end of the pandemic, the economy has started to improve. However, the continued impact of the later stages of the pandemic on the economy will require continued attention.



THANK YOU